OJAS PAL

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate building agentic AI, RAG, and machine learning applications. Develops verified LLM workflows, secure APIs, and cloud-deployed systems using Python and Google Cloud.

EDUCATION

University School of Automation and Robotics (USAR), GGSIPU East Campus, Delhi **2024 – 2028**
B.Tech in Artificial Intelligence & Data Science (AI-DS)

TECHNICAL SKILLS

Languages: Python, C++
GenAI & Agents: LangGraph, LangChain, RAG, Gemini API, Tool Calling, Human-in-the-Loop
Machine Learning: scikit-learn, PyTorch, Pandas, NumPy
Cloud & Data: Pinecone, Cloud Firestore, Firebase Auth, Cloud Run, Secret Manager
Backend & Apps: FastAPI, REST APIs, Streamlit
Developer Tools: Docker, Git, GitHub

PROJECTS

GitResolve: Autonomous GitHub Issue Repair Agent | GitHub **November 2026**
Python, LangGraph, FastAPI, Gemini API, Docker, Firestore, Cloud Run, Compute Engine
- Built a 13-stage AI workflow to investigate GitHub issues, reproduce bugs, verify fixes, and prepare draft pull requests for human review.
- Added 2 approval gates and repair budgets (50 model calls, 6 attempts); verified patches with pytest in network-isolated Docker and SHA-256 diff locking.
- Deployed the Vercel frontend, Cloud Run API, and GCE worker, coordinated through Firestore-backed asynchronous task state.

ML-DL Portfolio: 11 Models in One Streamlit App | GitHub **October 2026**
Python, Streamlit, scikit-learn, PyTorch, CatBoost, EfficientNet-B0
- Unified 11 ML/DL apps spanning prediction, recommendation, NLP, and vision into one Streamlit interface with cached pretrained models.
- Replaced an ~8 GB similarity matrix for 32,833 songs with on-demand cosine similarity to operate within a 2.7 GB RAM limit.
- Fine-tuned EfficientNet-B0 in 2 stages on 46 pet breeds from two public datasets, achieving 93.6% validation accuracy.

Medical RAG Chatbot | GitHub **September 2026**
Python, LangChain, Groq API, Pinecone, Hugging Face, Cloud Run
- Built a medical-reference Q&A app grounding LLM responses in relevant encyclopedia passages retrieved from Pinecone via LangChain.
- Indexed 5,800+ chunks from a 637-page source with 384-dimensional BGE embeddings and top-2 semantic retrieval.
- Deployed a containerized service on Cloud Run; lazy-loaded embedding models to resolve startup failures and improve readiness.

Personal Gemini Journal | GitHub **September 2026**
Gemini API, Firebase Auth, Firestore, Cloud Run, Secret Manager
- Developed a multi-user AI journal with contextual memory search combining Gemini embeddings and keyword scoring.
- Protected 9 REST endpoints with Firebase token and revocation checks; isolated user data under /users/{uid}.
- Managed Gemini keys with Secret Manager and least-privilege IAM; reused the last 3 entries as context without extra generation calls.

CERTIFICATIONS

- Google Cloud Gen AI Academy, APAC Edition, Cohort 3 (Hack2Skill)
- IBM SkillsBuild AI Automation & Intelligent Solutions Internship